

Laboratory session 1

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25. September 2026

1 Laboratory session 1

The first experiment is using standing waves inside a long tube in order to measure the speed of sound inside the gas the tube is filled with. The tube and experiment settings are specified by the following data:

Tube length: $L = 1244 \pm 1.5 \text{ mm}$.

input signal: $V_0 = 35 \text{ mV}$

Tubes:

K1: argon or CO_2 at $T = T_{\text{room}}$

K2: air at $T = T_{\text{room}}$

K3: air at $T = 70^\circ\text{C} = 343.15 \text{ K}$

K4: air at $T = 50^\circ\text{C} = 323.15$

$T_{\text{room}} = 21^\circ\text{C} = 294.15$

We analyzed the properties of the gas in tube K4. The resistance of the tube is measured to be $R = (34.55 \pm 0.01) \text{ k}\Omega$.

2 Tasks

2.1 Measurement of ν_k

We measured the following resonance frequencies ν_k between 200 Hz and 2 kHz :

K4: $\nu_k = [313 \pm 1, 450 \pm 1, 594 \pm 1, 740 \pm 1, 886 \pm 1, 1033 \pm 1, 1178 \pm 1, 1326 \pm 1, 1473 \pm 1, 1620 \pm 1, 1767 \pm 1, 1913 \pm 1]$

The other groups measured:

K3: $\nu_k = [301 \pm 1, 459 \pm 1, 607 \pm 1, 757 \pm 1, 907 \pm 1, 1057 \pm 1, 1208 \pm 1, 1359 \pm 1, 1510 \pm 1, 1661 \pm 1, 1813 \pm 1, 1963 \pm 1]$

K2: $\nu_k = [286 \pm 1, 419 \pm 1, 470 \pm 1, 563 \pm 1, 700 \pm 1, 840 \pm 1, 979 \pm 1, 1117 \pm 1, 1258 \pm 1, 1397 \pm 1, 1537 \pm 1, 1677 \pm 1, 1816 \pm 1, 1955 \pm 1]$

K1: argon: $\nu_k = [265 \pm 1, 393 \pm 1, 521 \pm 1, 650 \pm 1, 778 \pm 1, 905 \pm 1, 1035 \pm 1, 1164 \pm 1, 1293 \pm 1, 1423 \pm 1, 1551 \pm 1, 1680 \pm 1, 1810 \pm 1, 1938 \pm 1]$

CO_2 : $\nu_k = [225 \pm 1, 440 \pm 1, 544 \pm 1, 656 \pm 1, 764 \pm 1, 872 \pm 1, 979 \pm 1, 1091 \pm 1, 1199 \pm 1, 1209 \pm 1, 1418 \pm 1, 1527 \pm 1, 1635 \pm 1, 1745 \pm 1, 1854 \pm 1, 1963 \pm 1]$

2.2 Estimated error

We estimated our error to be around 1 Hz .

2.3 The best linear fit

In order to find the speed of sound in the gas, we want to use the equation $\nu_n = an + b$. However, we do not know which resonance frequency we are measuring, so we do not know which n belongs to which measured ν_n . This is why we set the reference of the measured ν_n to the first one measured. Therefore we get:

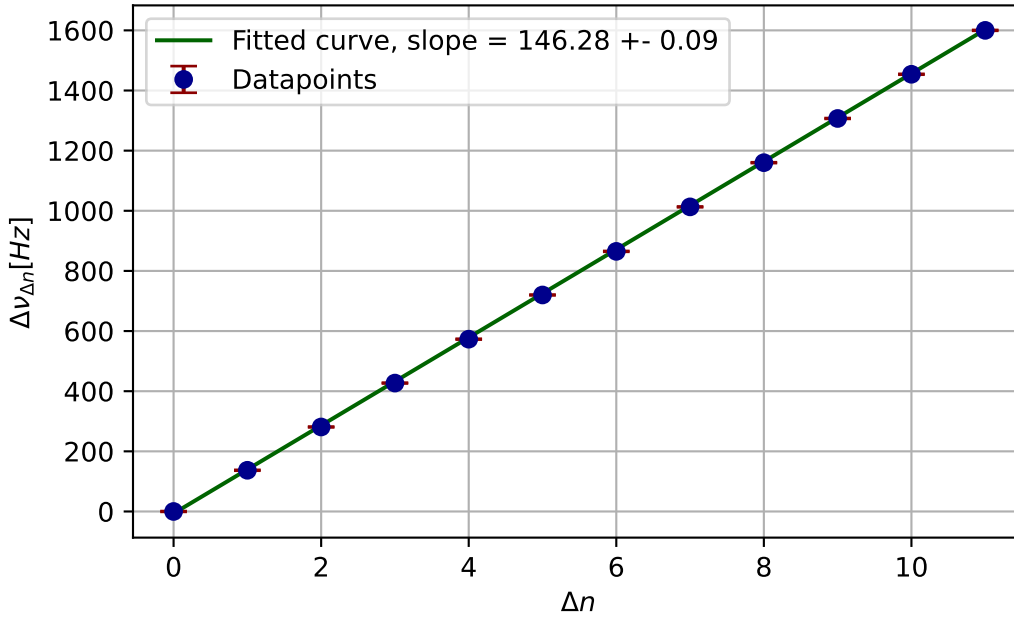
$$\nu_n - \nu_{n_0} = \Delta\nu_{\Delta n} = a\Delta n + b - (a(n_0) + b) = a(n - n_0) + b - b = a\Delta n = \frac{c}{2L}\Delta n$$

So the central relation is:

$$\Delta\nu_{\Delta n} = \frac{c}{2L}\Delta n$$

where Δn starts at 0 for the first measurement and for k measurements continues until $k - 1$ for the k^{th} resonance frequency. $\Delta\nu_{\Delta n}$ against Δn is plotted and evaluated in the following:

K4:

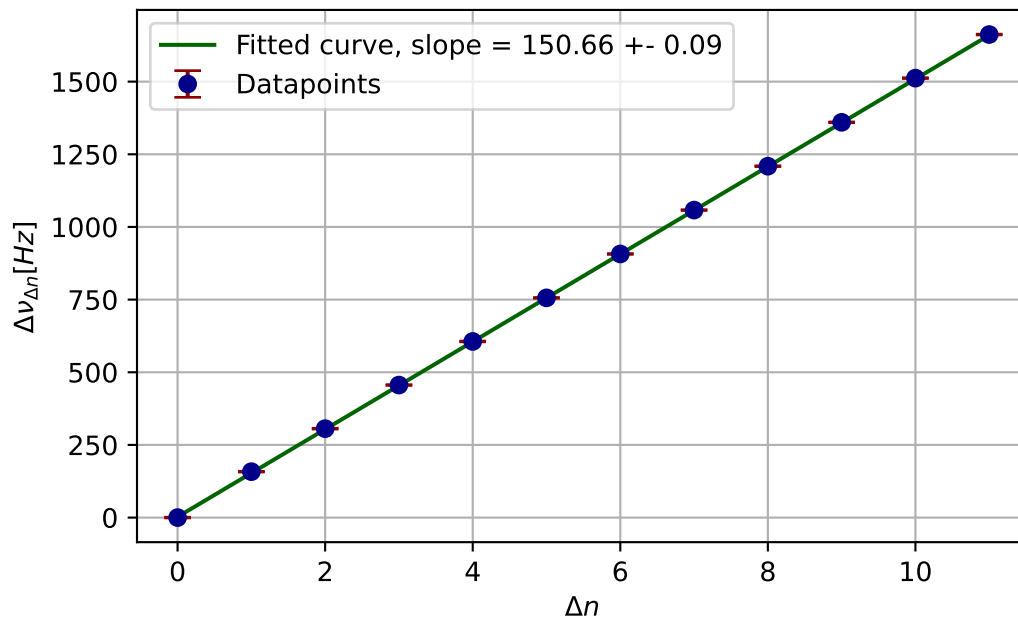


Given the slope of the fit curve, the velocity of sound can be calculated:

$$\text{speed of sound} = 363.95 \pm 0.49 \text{ m/s}$$

K3:

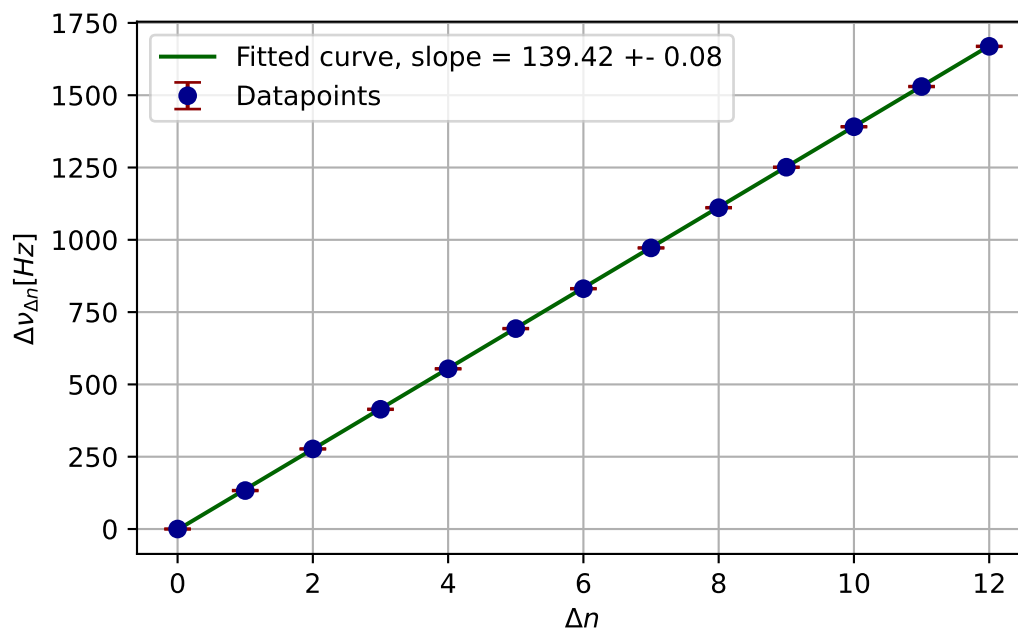
$$\text{The resistance of the tube is measured to be } R = (15.66 \pm 0.02)k\Omega$$



Given the slope of the fit curve, the velocity of sound can be calculated:

speed of sound = 374.85 ± 0.50 m/s

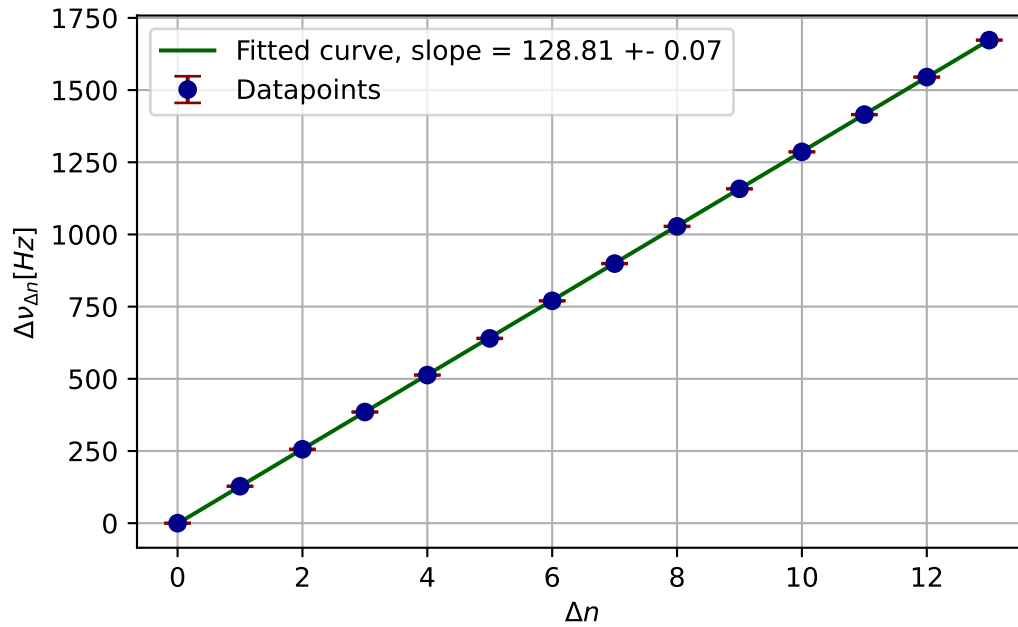
K2:



Given the slope of the fit curve, the velocity of sound can be calculated:

speed of sound = 346.87 ± 0.46 m/s

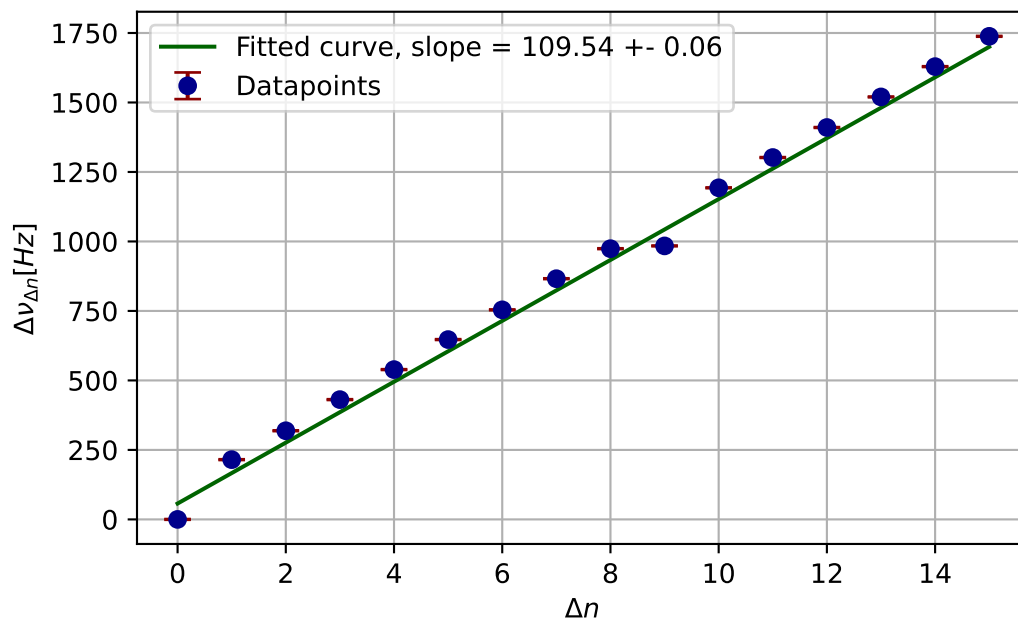
K1, argon:



Given the slope of the fit curve, the velocity of sound can be calculated:

speed of sound = 320.47 ± 0.42 m/s

K1, CO_2 :



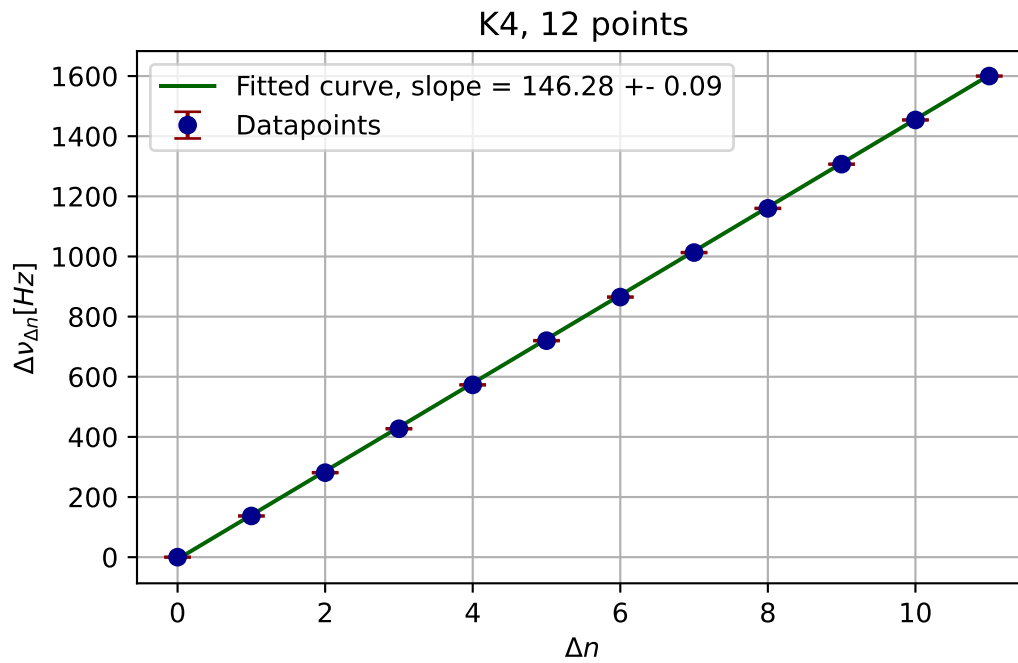
Given the slope of the fit curve, the velocity of sound can be calculated:

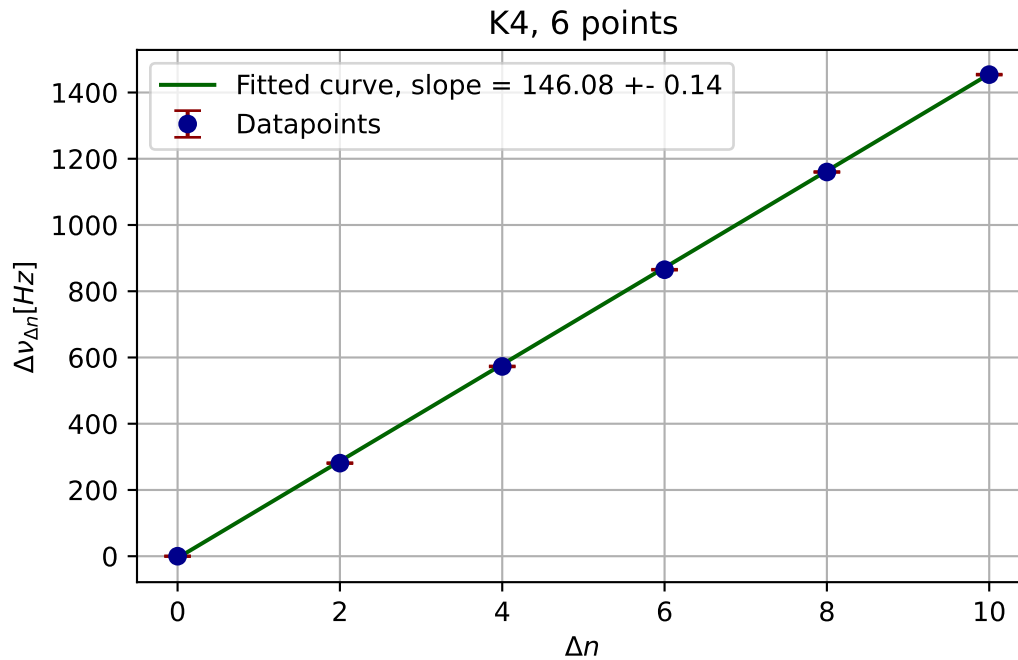
speed of sound = 272.53 ± 0.36 m/s

2.4 Most important contribution to uncertainty

The most important contribution to the error is the inaccuracy in finetuning the resonance frequency. We estimated our accuracy in doing so to $1Hz$.

The uncertainty shrinks the more points are included in the fit. In the following we see the plot for K4, once with 12 points and once with 6:





As we can see, the uncertainty on the slope is more than 50% greater if we only choose half of the points.

2.5 Degrees of freedom

For an ideal gas, the velocity of sound is given by

$$c(T) = \sqrt{\frac{(f + 2)RT}{fM_{mol}}}$$

where f are the degrees of freedom, R is the gas constant, T the temperature of the gas and M_{mol} the molar mass. This equation can be solved for f :

$$f = \frac{2RT}{c^2 M_{mol} - RT}$$

We calculate for f :

K4:

The calculated amount of degrees of freedom from the experiment is 5.81 ± 0.09

K3:

The calculated amount of degrees of freedom from the experiment is 3.89 ± 0.05

K2:

The calculated amount of degrees of freedom from the experiment is 4.71 ± 0.07

K1, argon:

The calculated amount of degrees of freedom from the experiment is 2.94 ± 0.03

K1, co2:

The calculated amount of degrees of freedom from the experiment is 5.95 ± 0.10

In theory, the air inside K2, K3, K4 should have 5 degrees of freedom. These consist of 3 for translation and 2 for rotation. CO2 can also be described as a 2-atom molecule because it is linear, therefore it should also have 5 degrees of freedom. Argon is a one-atom gas and only has 3 translational degrees of freedom. ## Comparison to simulation and theory

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3 Simulation

3.1 Phase diagram

We change the parameters in the simulation and observe how the simulation changes. Therefore, we double the initial values and plot the simulation's output.

As seen in the plots above, the output still is linear and the starting temperature blabla. The density $\rho^* = 0.1$ might have been a bit too high and therefore the Lennard-Jones gas does not act as an ideal gas.

...

3.2 C_V , Z and degrees of freedom of a simulated Lennard-Jones gas

We used a Lennard-Jones gas with the following properties: energy flux 5, particle number $N = 5000$, temperature $T^* = 1.4$, density $\rho^* = 0.1$, pressure $p = 0.01$ and a timestep of $t = 0.005s$

$$C_V = \frac{dE \cdot t}{\Delta T}$$

$Z = \frac{pV}{NT}$ For an ideal gas the compressibility factor Z is equal to 1. There is no Boltzmann constant because we use reduced Lennard-Jones units. As seen in our simulation (hier verlinken oder so) $Z = ?$. This means our gas can be described as an ideal gas as Z is close to 1.

$$\text{degrees of freedom } f = \frac{2c_V}{R}$$

We expect the gas to have 3 degrees of freedom because it's a one-atom gas and then we only have the translational degrees of freedom. This fits well to the measurement. ...

3.3 Different temperatures at fixed gas density

triple point temperature: $T_{*t} = 0.69$ critical temperature: $T_{*c} = 1.32$

We chose the following temperatures between the triple point temperature and ten times the critical temperature: $T \in [0.69, 13.2]$

C_V and Z change/do not change with temperature. ...

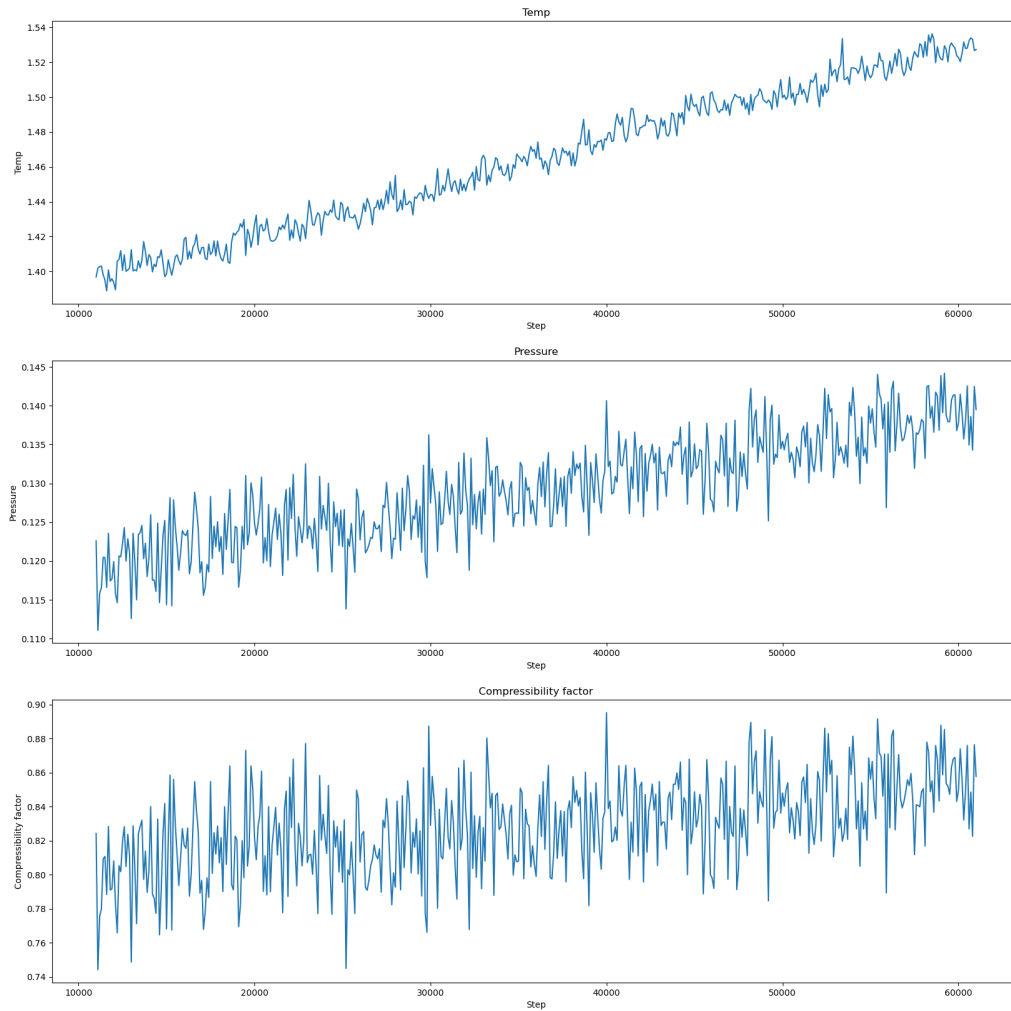


Abbildung 1: simulation

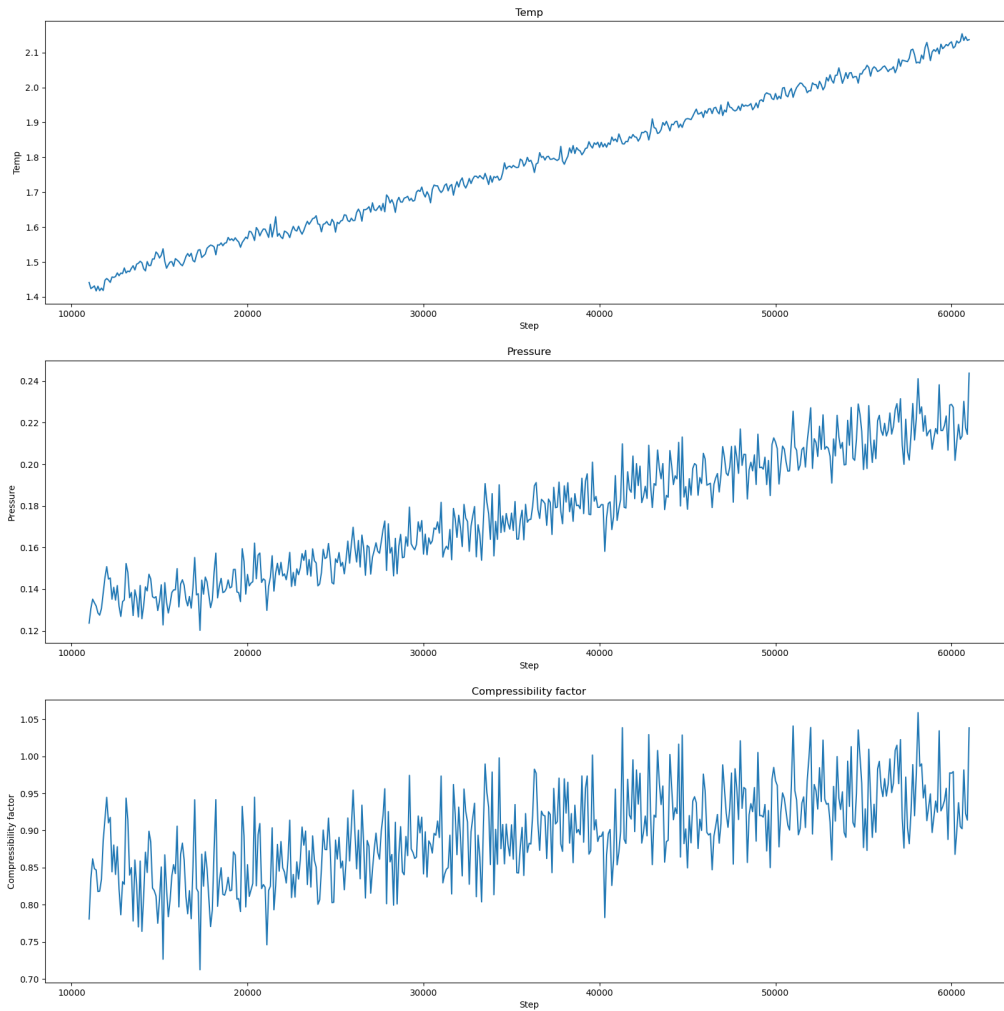


Abbildung 2: simulation with doubled number of particles

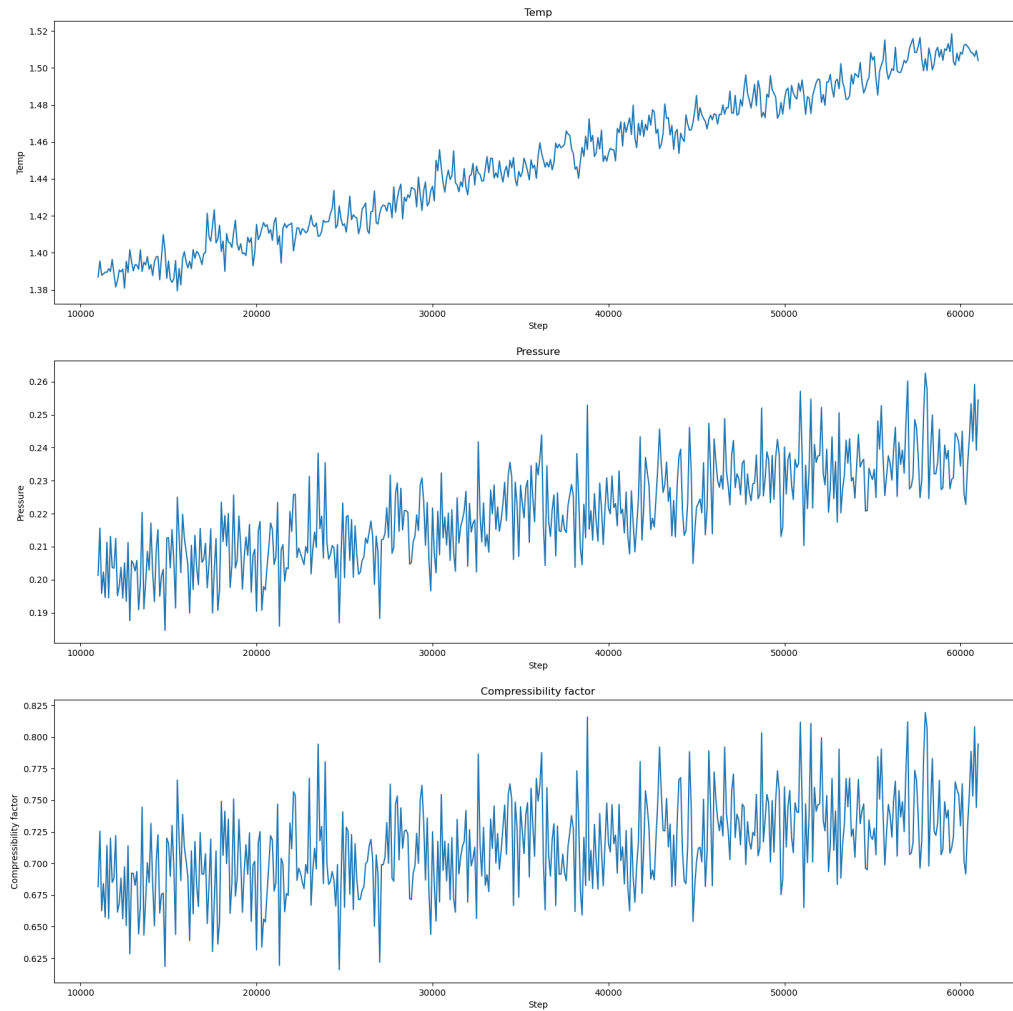


Abbildung 3: simulation with doubled density

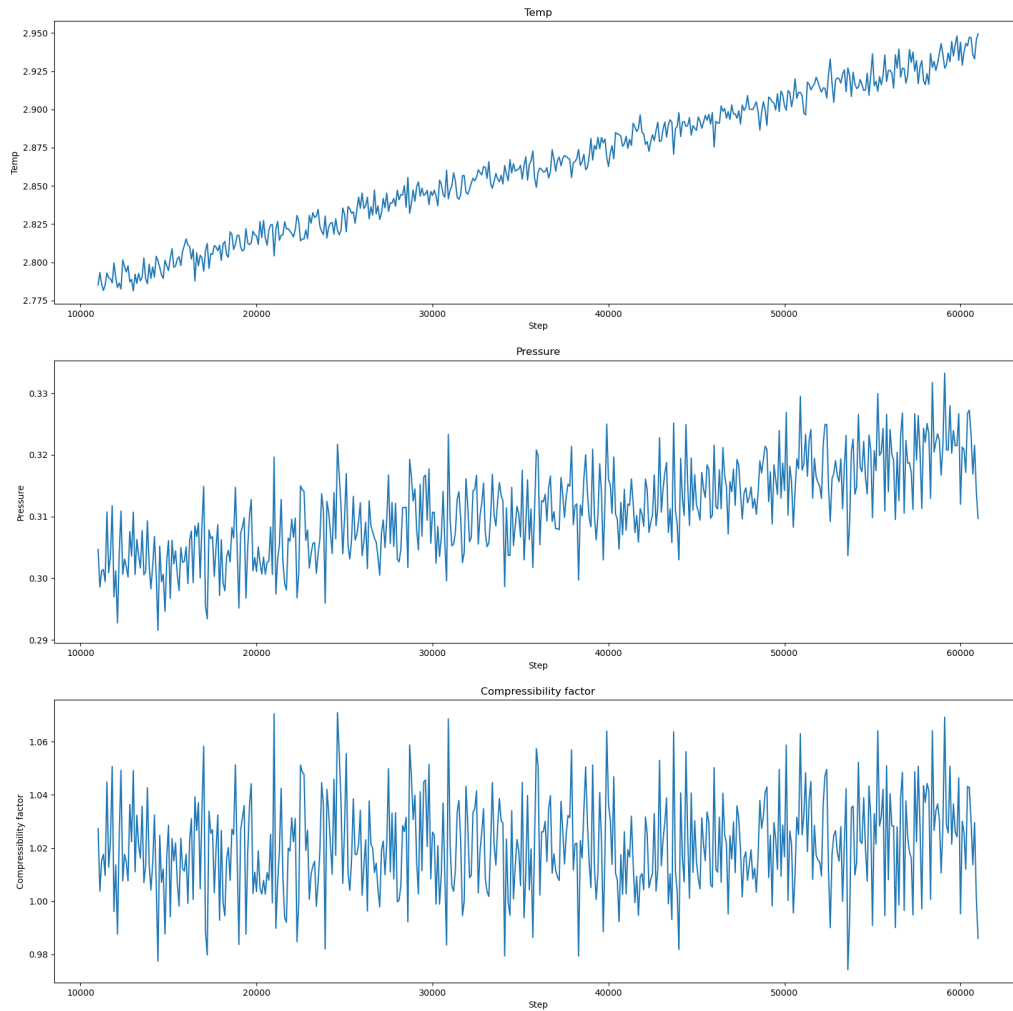


Abbildung 4: simulation with doubled starting temperature

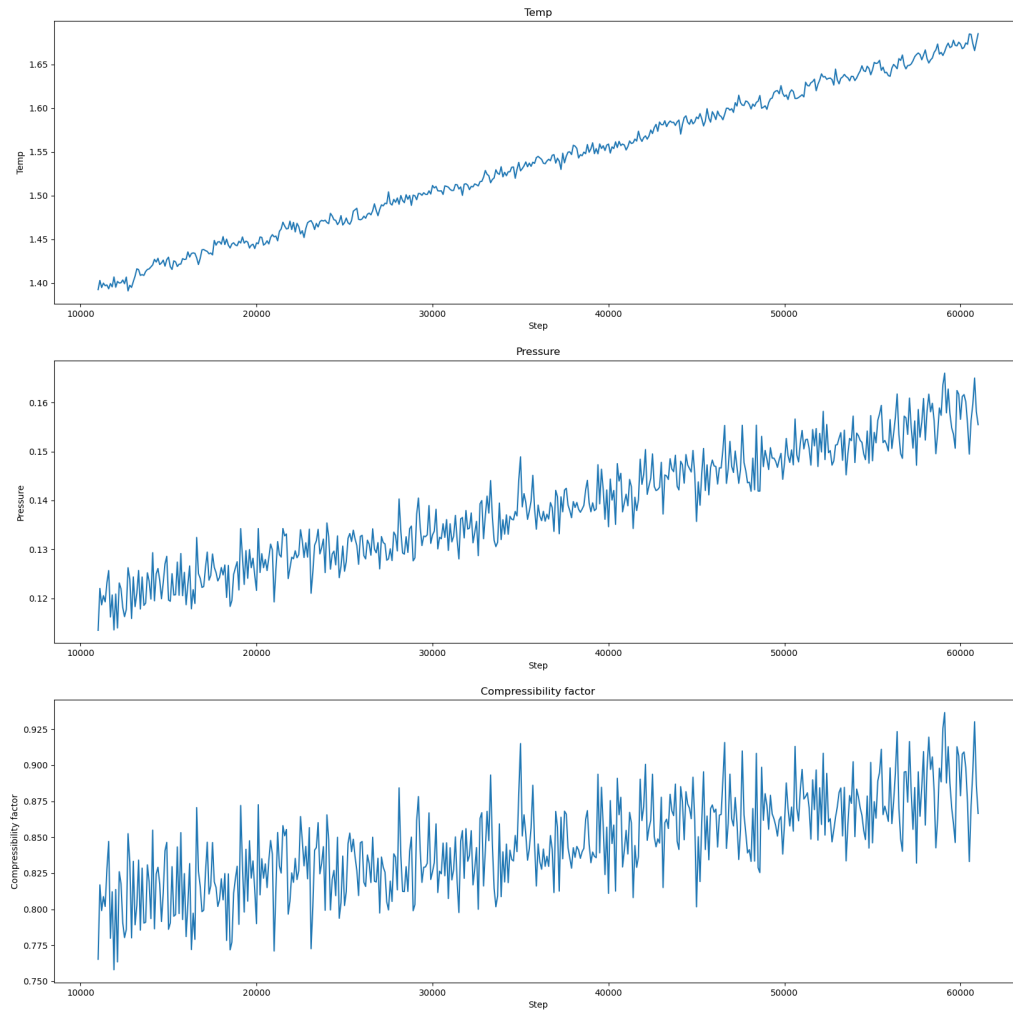


Abbildung 5: simulation with doubled energy flux

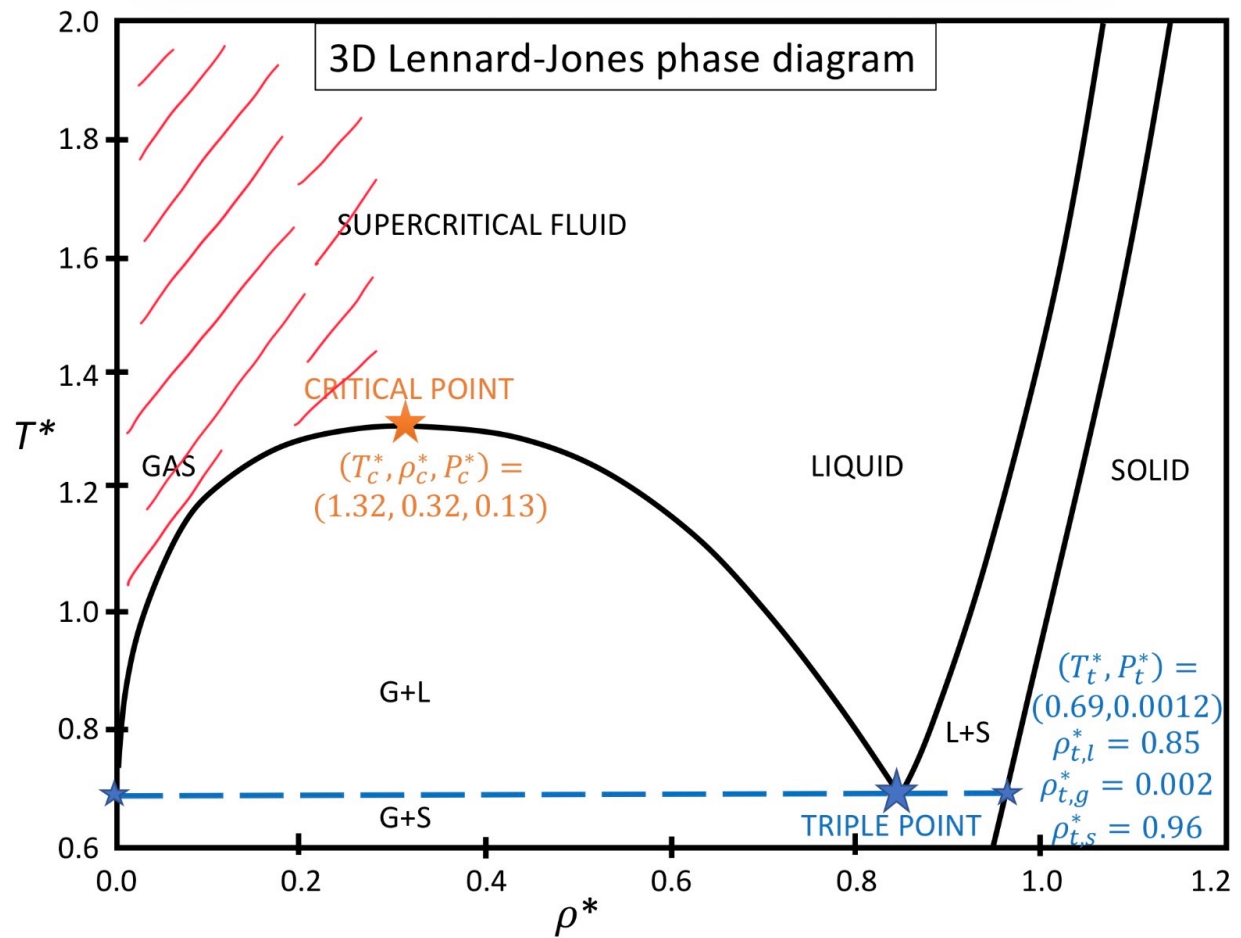


Abbildung 6: Lennard Jones phase diagram: the marked area is the area where our simulations take place

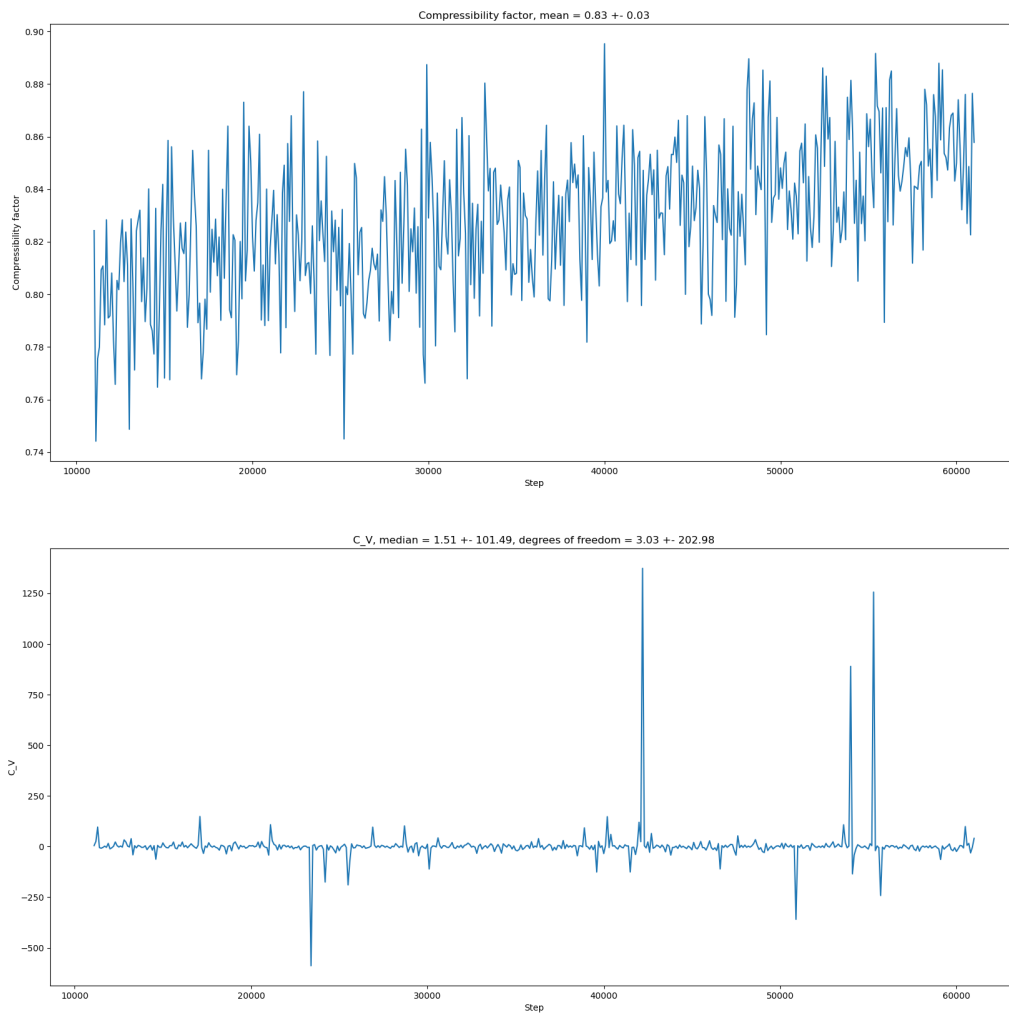


Abbildung 7: Z and C_V

3.4 Temperatures above the critical point

For a temperature T^* we used the following densities: $\rho = \text{hier Dichtenrein}$
 C_V and Z change/do not change with density ...

3.5 Determination of C_V

The N2 models have ? degrees of freedom according to the simulation.

The number of degrees of freedom derived from sound velocity measurements in the lab is $f_{lab} = ?$

We expected f for rigid N2 to be 5? and for N2 with vibrations to be 7? ...